Review

Clock Synchronization for Networked Robotic Sensor Systems from Packet Timestamps to Acquisition Events

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Abstract

Networked robots depend on temporally consistent camera, inertial and ranging measurements, yet agreement between network clocks does not establish agreement between physical acquisition events. This critical review examines that gap across packet synchronization, local clock mapping, sensor triggering and temporal calibration. A targeted literature search and source-level evidence audit support a classification organized around the boundaries at which time is distributed, translated, attached to measurements and consumed. The comparison shows why hardware timestamping reduces selected transport uncertainties while leaving path asymmetry, sensor filtering, exposure timing and clock-domain translation to be addressed separately. Representative published results are compared with their measurement boundaries and experimental conditions retained, rather than pooled into an accuracy ranking. An explicit error budget connects acquisition requirements to residual drift, synchronization intervals and communication cost through reproducible illustrative calculations. The synthesis supports a layered design in which network timing, event scheduling and residual temporal calibration have distinct responsibilities, with timing validity exposed during loss and recovery of synchronization. Public robotic datasets provide useful acquisition examples but do not replace network timing traces. A reporting checklist identifies the evidence needed to assess synchronization claims across heterogeneous robotic sensor networks.

Keywords: clock synchronization; sensor networks; robotic sensing; hardware timestamping; precision time protocol; clock drift; temporal calibration; communication overhead

1. Introduction

A multi-camera robot can produce images carrying almost identical timestamps while observing different physical instants. Conversely, two measurements can be acquired simultaneously and delivered to a fusion process at very different times. These situations have different causes and require different remedies. Sensor-network synchronization research has long distinguished clock agreement, energy consumption and scalability in wireless sensor networks (WSNs) [1]. Packet-network and wireless local area network (WLAN) surveys further establish that timestamp placement, uncertain delays and oscillator behavior affect the attainable agreement [2,3]. For robotic sensing, the additional question is whether the time attached to a measurement identifies the physical event that the estimator assumes it identifies.

Existing reviews already provide substantial coverage of this subject's components. Puttnies et al. compare time estimators across wired and wireless settings with explicit attention to experimental conditions [4]. Dang et al. classify Industrial Internet synchronization mechanisms and requirements [5]. The 2026 survey by Seliem et al. extends wireless coverage across technologies, architectures, uncertainty models and deployment

classes [6]. Robotic calibration reviews address how cameras, light detection and ranging sensors (LiDARs), and inertial measurement units (IMUs) can be spatially and temporally registered [7]. Cross-layer wireless sensor-network modeling is also established within this journal [8]. A further generic protocol taxonomy would therefore add limited value.

This review instead follows the timing chain from a network reference to an acquisition event and finally to its timestamp at the consumer. Its scope is heterogeneous robotic sensor networks, including sensors on one robot connected through several interfaces and sensing nodes distributed across robots. Cameras provide the principal application example because exposure start, exposure midpoint and delivery time are visibly different. IMUs and LiDARs reveal corresponding problems with filtering, sampling phases and scan timing. The analysis remains centered on network architecture, communication paths, clock domains and the cost of maintaining timing information. Image reconstruction, general simultaneous localization and mapping (SLAM), and embodied intelligence are considered only where they clarify a timing requirement or an evidence boundary.

Four contributions structure the synthesis. First, the review distinguishes network clock distribution, local clock mapping, acquisition scheduling and estimator-side registration, then identifies what evidence connects these stages. Second, it compares representative synchronization results while preserving the reported metric and measurement boundary. Third, it develops a common error budget and reproducible calculations that connect residual drift and application tolerance to update intervals and protocol traffic. These calculations are design illustrations, not new measurements or validated performance guarantees. Fourth, it proposes an acquisition-focused reporting and verification checklist, including failure and recovery behavior. The intended contribution is an engineering synthesis of established mechanisms, not a new synchronization protocol or an unqualified claim of first coverage.

2. Review Method and Evidence Boundaries

2.1. Search and selection

The review used targeted, iterative searches of publicly accessible scholarly and technical material, last checked on 19 October 2026. Web searches located publisher records, author repositories, arXiv papers and official standards or implementation documentation. Representative queries included “clock synchronization survey robotic sensor networks camera PTP review 2024 2025 2026”, “TICSync clock synchronization”, and “VersaVIS synchronization 2020”. Specific-title queries located hardware sensor-synchronization and cross-vendor IEEE 802.1AS studies. Focused searches checked recent surveys and the status of IEEE 1588 and IEEE 802.1AS. Crossref title queries and DOI lookups were used for bibliographic verification. The supplementary search record preserves the actual queries, retrieved records and access failures.

Sources were selected for explaining a timing mechanism, identifying a limit or failure mode, documenting an implementable interface, or providing a robotic acquisition example. Foundational protocols were retained even when older than recent application papers. Sources discussing synchronization of robot actions without addressing physical clock agreement were treated separately. Manuscripts, standards and documentation were not assumed to offer interchangeable evidence. Selection was purposive and refined during reading; it was not preregistered. No subscription database export, exhaustive screening denominator, independent dual screening or PRISMA study flow is claimed. This is a critical narrative review with an auditable evidence ledger, not a systematic review or a meta-analysis.

2.2. Extraction and interpretation

Each source was recorded with a persistent identifier or primary URL, access date, accessible evidence location, supported claim and limitation. Bibliographic verification was separated from verification of a scientific statement. A real DOI establishes the existence of a paper; it does not establish that a particular performance claim appears in it. Full papers were read where obtainable. Where only a publisher abstract, indexed primary text or a public standards summary was accessible, the use of that source was restricted accordingly. Experimental values in Table 4 come from accessible primary full texts, rather than from secondary surveys. Technical documents support interface semantics and configuration requirements, not independent performance claims.

For quantitative comparisons, the extraction retained the measured quantity, timestamp boundary, statistic, hardware or medium, and operating conditions. Unavailable details were recorded as not reported or not verified. A small signed mean, a standard deviation, a root-mean-square error and a maximum absolute error were not converted into one another. Neither were internal servo residuals treated as independent measurements of external timing accuracy. Repeated calibration offsets were distinguished from direct event-time measurements. Numerical examples introduced by this review are labeled as calculations and can be regenerated from the accompanying code. No raw public dataset was downloaded for reanalysis, and no new experimental sensor or network results are reported.

Table 1 positions the review relative to adjacent work. It identifies differences in organizing question; it does not assert that these earlier sources never discuss acquisition or implementation details.

Table 1. Relationship to existing reviews and a directly relevant JSAN study. Descriptions reflect the accessible scope of each source; the present review's contribution is the connection across measurement boundaries.

Source	Main organizing question	Relationship to this review
Sundararaman et al. [1]	How can resource-constrained WSN clocks be synchronized?	Foundational protocol and deployment distinctions
Lévesque and Tipper [2]; Mahmood et al. [3]	How do packet networks and WLANs distribute time?	Transport impairments and timestamp location
Puttnies et al. [4]	Which estimators achieve which results under which conditions?	Basis for preserving conditions in comparisons
Dang et al. [5]; Seliem et al. [6]	How do industrial and wireless synchronization mechanisms differ?	Broad network taxonomies; this review narrows to robotic acquisition
Lyu et al. [7]	How can robotic sensor modalities be spatiotemporally calibrated?	Estimator-side registration and calibration context
Brunner et al. [9]	Why can cross-vendor generalized PTP systems fail to interoperate?	Primary evidence for clock jumps, delay estimation and configuration failure
Present review	What connects a synchronized network clock to a valid acquisition timestamp?	Cross-boundary error accounting, traffic trade-offs and verification requirements

3. A Framework from Network Time to Acquisition Time

3.1. Four responsibilities in the timing chain

The timing chain has four responsibilities. Network distribution establishes a relation between clocks at communicating nodes. Local mapping transfers that relation between a network interface, an embedded timer and the clock exposed to an application.

Acquisition scheduling or event capture establishes when a physical measurement occurs relative to those clocks. Temporal registration reconciles remaining offsets with the measurement model used by the estimator. Figure 1 separates these responsibilities and the distinct path by which data reach the consumer. A network interface controller (NIC), a microcontroller unit (MCU), and a host can each expose a separate clock. Network distribution may use the Precision Time Protocol (PTP) or generalized Precision Time Protocol (gPTP).

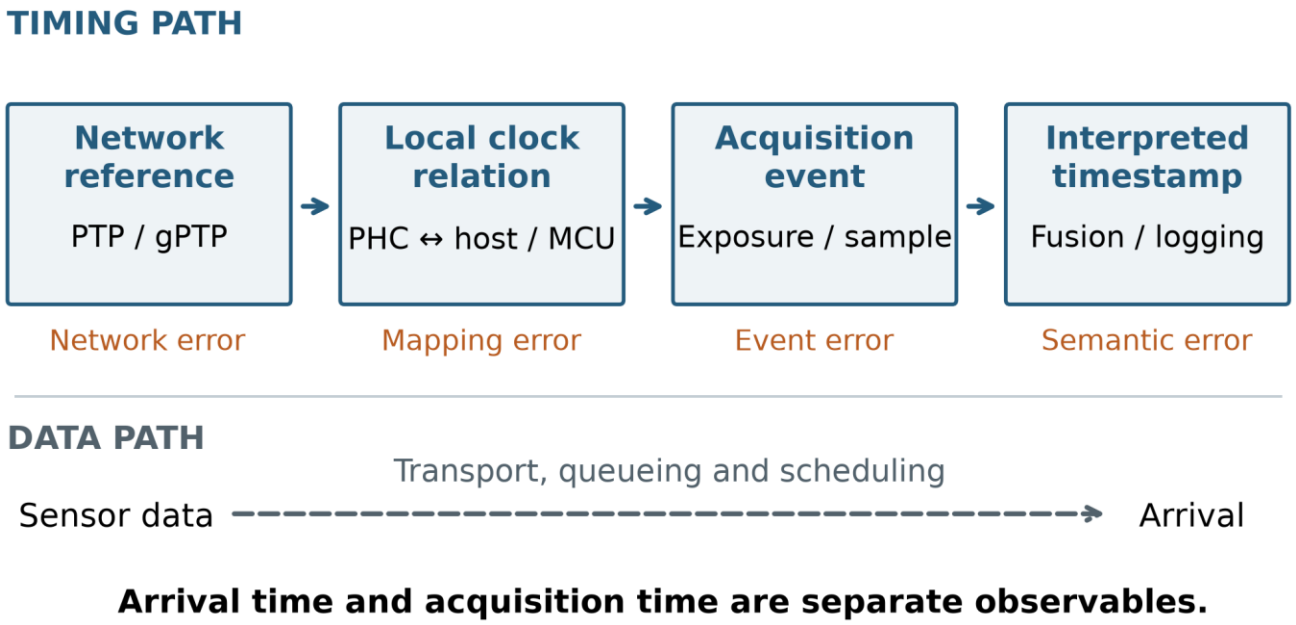


Figure 1. Timing and data paths in a heterogeneous robotic sensor network. Network clock agreement, local clock translation, physical event timing and timestamp interpretation require separate evidence. Data may arrive late without corrupting acquisition time when a valid source timestamp and its mapping are retained. The diagram is an original conceptual synthesis; it does not depict an experimental system.

A local clock can be approximated over a specified interval as

$$C_i(t) = a_i t + b_i + w_i(t). \quad (1)$$

Here, t is reference time, a_i is the dimensionless rate, b_i is the offset in seconds, and $w_i(t)$ represents departures from the affine approximation. Temperature changes and other oscillator effects restrict the interval over which constant rate is adequate. The affine model is useful for separating frequency and offset estimation, but it should not be treated as an exact lifetime model [10].

Network-clock agreement alone does not close the chain. Linux distinguishes scheduler, software and hardware packet timestamps, including raw hardware-clock timestamps [11]. The LinuxPTP tools separately address synchronization through the network and synchronization between a PTP hardware clock (PHC) and the system clock [12,13]. Thus, a correctly disciplined interface clock can coexist with an incorrectly interpreted application timestamp. The problem becomes more pronounced when a microcontroller or USB-connected sensor maintains another independent clock.

The framework is orthogonal to protocol names. PTP can be implemented with different timestamping support, while a hardware-triggered camera can still require software clock translation. Temporal calibration can operate on data whose node clocks are already synchronized. Describing a system simply as “hardware synchronized” or “software synchronized” suppresses the distinction that determines its residual error.

3.2. Errors and observability

For a four-timestamp exchange, suppose clock rates have been expressed in a common unit and offset is approximately constant during the exchange. Let t_1 and t_4 be reference-side transmit and receive times and t_2 and t_3 be receiving-node times. If forward and reverse delays are d_f and d_r, the conventional symmetric-delay estimate satisfies

$$\hat{\theta} = [(t_2 - t_1) - (t_4 - t_3)]/2 = \theta + (d_f - d_r)/2 + \epsilon. \quad (2)$$

The term ϵ collects timestamp noise and residual model error. Equation (2), consistent with the NTP four-timestamp construction, makes the asymmetry bias explicit [14]. More frequent exchange can improve tracking or reduce selected random errors; it cannot identify an arbitrary constant asymmetry from these observations alone. The impossibility results of Freris et al. establish the broader offset-delay ambiguity under their stated clock and delay assumptions [10]. Additional calibration, known path structure or independent timing observations can reduce that ambiguity. The result therefore limits the information available from timestamps; it does not prohibit well-engineered synchronization.

For a reported acquisition timestamp, the residual error can be organized as

$$e_i = e_{net,i} + e_{map,i} + e_{event,i} + e_{sem,i}. \quad (3)$$

The terms denote network distribution error, local clock mapping error, physical event capture or scheduling error, and timestamp-semantic error. This is an accounting decomposition defined for this review, not an assumption that these components are independent or directly observable. The boundary between terms must be declared to prevent double counting: for example, a calibrated fixed sensor delay should be assigned once. For conservative component bounds, U_i is at most the sum of the component bounds, and the pairwise bound between sensors i and j is $U_i + U_j$. Correlated errors may cancel in a relative measurement or reinforce in a particular comparison; a root-sum-square combination requires a justified covariance model.

Two timestamps agreeing to a nanosecond can consequently coexist with millisecond uncertainty in the sampled event. This is a logical consequence of the chain, not a general claim about particular hardware. Equally, a common error in all node clocks can preserve relative acquisition alignment while preventing traceability to an external timescale. Relative and absolute accuracy must therefore be reported separately.

Table 2. Mechanisms classified by responsibility in the timing chain. The final column identifies evidence still needed after the mechanism operates successfully.

Mechanism	Principal responsibility	Residual evidence required
NTP or software-timestamped PTP	Distribute reference time through packet exchange	Path asymmetry, host timing boundary, event semantics
Hardware-timestamped PTP or gPTP	Observe packet events close to the network interface	Local clock translation and sensor acquisition timing
Embedded timer and trigger wiring	Schedule or capture acquisition events	Cable/interface delays, internal sensor behavior, network reference mapping
Affine clock translation	Map sensor and host clocks	Delay assumptions, model validity and behavior after resets
Motion-based temporal calibration	Register sensor observations in an estimator	Observability, time variation, estimator-model mismatch
Message timestamp matching	Associate delivered measurements	Correctness of upstream clocks and acquisition timestamps

4. Packet Synchronization and the Network Contribution

4.1. Timestamp location and clock-aware infrastructure

IEEE 1588-2019 defines a configurable packet-clock synchronization protocol with application profiles [15]. Its public description identifies the potential for high accuracy in an appropriately designed system. That potential is not a device-independent guarantee. Hardware timestamps move the observation boundary toward the network interface and remove selected operating-system and queueing uncertainties from the interval being measured. They do not automatically remove physical-link asymmetry or all delays between the timestamping location and the medium. Linux's timestamping interfaces make these distinctions concrete: timestamps generated before the scheduler, by software near the driver, and by hardware correspond to different events [11].

Clock-aware intermediate nodes can account for residence time or divide a timing path into controlled segments. Their benefit depends on compatible protocol profiles, correction behavior and timestamp calibration. IEEE 802.1AS-2020 is the edition used by many reviewed implementations [16]. At the review date, IEEE's official record identifies IEEE 802.1AS-2025 as its successor, published on 17 December 2025 [17]. The older edition is therefore cited as a historical implementation baseline. Detailed conclusions about the revised normative clauses cannot be drawn from the public summary alone.

The cross-vendor study by Brunner et al. illustrates why implementation evidence matters. It reports problems associated with mean-link-delay computation, adjustment of a clock used for timestamping, and inconsistent traffic-class assignment [9]. A local time jump can contaminate rate and delay estimation at neighboring devices, rather than remaining confined to the device being corrected. Their results support reporting startup and reconfiguration behavior alongside steady-state offset. They do not establish that all products have those defects or that the same defects persist unchanged in the 2025 edition.

White Rabbit illustrates a different design point: specialized Ethernet timing resources can combine time and frequency distribution with careful calibration [18,19]. Its relevance is architectural. A robot using commodity interfaces does not inherit the performance of calibrated timing infrastructure merely by using Ethernet or PTP. The practical question is which hardware, link calibration and deployment constraints can be maintained on the target sensor network.

4.2. Wireless broadcast and multi-hop propagation

Wireless synchronization is affected both by timestamp implementation and by access to a shared medium. Reference-Broadcast Synchronization (RBS) uses a common reception event to compare receiver clocks, excluding sender-side uncertainty from that comparison [20]. The benefit depends on what is shared: comparable reception events do not remove receiver-specific delays or establish a sensor's exposure time. Timing-sync Protocol for Sensor Networks (TPSN) provides a hierarchical sender-receiver design [21]. FTSP combines MAC-layer timestamps, flooding and skew estimation [22]. Their differences are best interpreted through timing observations and topology, rather than a simple ranking of algorithm names.

Propagation speed also matters. A node receiving an old reference must extrapolate it using a clock-rate estimate. PulseSync analyzes the joint influence of communication latency, drift and network diameter and uses rapid propagation to reduce the age of timing information [23]. Glossy integrates flooding and synchronization through closely timed IEEE 802.15.4 transmissions [24]. Its channel use and tight radio timing are part of the synchronization mechanism; its performance should not be transferred directly to Wi-Fi camera transport. Fast floods can reduce reference-age error while consuming short periods in which other communication or processing is constrained.

These observations connect clock synchronization to communication engineering. A low packet count can conceal long receiver-on periods, and a low transmission duty cycle can still require tight scheduling of network activity. Conversely, sharing a broadcast reference can amortize transmissions across many receivers. Scalability therefore concerns the number of nodes, the number of timing links, reference age and radio activity, not just the number of synchronization packets.

4.3. Hybrid wired and wireless paths

A hybrid network introduces clock and delay boundaries at the interfaces between technologies. Caleya-Sanchez et al. implement a 5G transparent-clock arrangement and examine PTP message rates using commercial timing equipment [25]. Their measurements show the feasibility of accurate clock transfer for the tested configuration. The full text states that the experiment considered PTP traffic only and measured offsets over 20-minute intervals. It also identifies traffic-load evaluation as future work. Thus, its results do not directly establish performance under concurrent multi-camera traffic, mobility or robot acquisition timing.

That distinction is consequential. A bridge can compensate a measured network residence time without observing a camera's integration interval. A node can recover the reference clock yet still need to translate it into a sensor timer. The appropriate comparison is therefore conditional: what path is timed, which clocks are observed, and which portion of the acquisition chain remains outside the experiment? Emerging in-network congestion marking addresses another portion of this problem by providing information about variable queueing to synchronization filters [26]. This recent work is a preprint and is treated as a research direction, not an independently established robotics result.

5. Sensor Acquisition and Software Responsibilities

5.1. Cameras, IMUs and scanning sensors

For a global-shutter image with exposure start t_0 and duration τ_{exp}, the exposure midpoint is $t_0 + \tau_{exp}/2$. A simplified rolling-shutter model adds the row-dependent delay r τ_{row}:

$$t_{cam,k}(r) = t_{0,k} + r\tau_{row} + \tau_{exp,k}/2. \quad (4)$$

Here, k indexes the frame and r the row; τ_{row} is zero for the global-shutter simplification. The model assumes a known row timing and integration interval. Motion blur and nonuniform illumination can make one effective timestamp an approximation even when the exposure interval is known. It follows that simultaneous frame-start triggers do not guarantee identical exposure midpoints if automatic exposure selects different durations.

VersaVIS explicitly addresses exposure compensation, sensor triggering and host-clock translation [27]. Its camera-IMU calibration also demonstrates why filtering matters: estimated offsets depend on the IMU filter setting. A data-ready interrupt or packet timestamp must therefore be interpreted relative to the sensor's filtering and sampling pipeline. Treating every fixed delay as random clock error makes calibration harder to understand; treating every delay as fixed ignores configuration-dependent and time-varying components.

The Matter of Time describes a field-programmable gate array (FPGA) architecture with a shared timer, hardware triggering and distinct sensor interfaces [28]. Recent hardware-based work similarly highlights the relationship between exposure control and IMU sampling [29]. Distributed acquisition research also distinguishes clock synchronization from sample synchronization [30]. These sources support a consistent principle: the event observed by the timestamp must be declared. They do not justify assuming that all cameras, IMUs or LiDARs expose the same triggering capabilities.

Scanning LiDAR adds another interval. A frame or scan timestamp is insufficient to describe all point acquisition times during motion; the appropriate model may need firing offsets or phase information. Ouster documentation separately describes timestamp-source modes, timer resolution and phase-lock settings [31]. Their separation is evidence that a synchronized timer and a synchronized scan phase are different configuration concerns. A timestamp's numerical resolution is also distinct from its accuracy relative to a reference.

Table 3. Acquisition timing distinctions that remain after network clock synchronization. The implications are deductions from the measurement models and cited implementation evidence, rather than new empirical measurements.

Event or timestamp	Ambiguity	Required information
Camera frame start	Unequal exposure durations imply unequal midpoints	Trigger event definition, exposure duration and shutter model
IMU data ready	Filtering or internal conversion can separate output availability from physical sampling	Filter configuration, sample-clock relation and documented group delay
LiDAR frame or scan	Points within a scan are acquired at different times	Per-point offsets or firing model and scan-phase convention
NIC receive timestamp	Packet arrival is later than sensor acquisition	Source timestamp and mapping between sensor and reference clocks
Host callback timestamp	Queueing and scheduling enter the timestamp	Driver timestamp provenance and clock domain

5.2. Clock mapping and temporal calibration

Software has several distinct roles. TICSync learns a mapping between distributed clocks incrementally and provides probabilistic error estimates under its modeling assumptions [32]. Such a mapping can preserve a sensor's native timestamps without continually stepping its clock. It remains sensitive to the information available in timestamp exchanges and to changes in the clock model. Reset detection and the validity interval of the map are therefore as important as a fitted offset.

Temporal calibration uses another kind of observation. Furgale et al. jointly estimate spatial and temporal registration in a continuous-time framework [33]. Li and Mourikis analyze online time-offset estimation and its identifiability, including degenerate motion cases [34]. Qin and Shen optimize temporal offset together with visual-inertial states [35]. Twist-n-Sync uses angular-rate measurements from rigidly connected gyroscopes as a shared temporal signal [36]. These methods can exploit physical observations that packet timing lacks. Their validity is conditional on sensor models, motion excitation and the form of temporal error represented in the estimator.

Calibration should consequently complement an explicit timing architecture. A constant-offset parameter can absorb a fixed acquisition delay, but cannot be assumed to represent arbitrary per-frame scheduling jitter. A time-varying estimator may track some changes without identifying their network or sensor origin. A good final trajectory does not uniquely validate each clock or timestamp boundary because other estimated quantities can compensate for temporal errors. Calibration quality and clock-transfer quality should be reported as related but separate results.

5.3. Middleware association, delivery and timescales

Message association is downstream of timestamp generation. The ROS 2 ApproximateTime implementation compares timestamps supplied with messages and selects compatible groups [37]. It does not measure oscillator drift, discipline device clocks or

recover an unobserved acquisition instant. Increasing an association tolerance can admit more message pairs while allowing a larger physical separation between their sampling events. The tolerance is an application decision, not a measured synchronization accuracy.

ROS clock abstractions also distinguish time domains and expose clock jumps [38]. Simulation time, wall time and a steady clock serve different purposes. In a distributed robot, system time and a PHC can additionally differ in timescale or offset; phc2sys documents handling of these relations [13]. A timestamp record should identify the source clock and conversion applied, rather than relying on a numeric epoch or unit alone.

Finally, communication latency and acquisition-time error must be kept separate. A delayed packet can carry an accurate event timestamp while arriving too late for a control deadline. A promptly delivered packet can carry a poor callback timestamp. Preserving source timestamps addresses the first timing question, while network scheduling, buffering and out-of-sequence estimation address the delivery question. Their interaction is important, but one metric cannot substitute for the other.

6. What Published Performance Results Actually Establish

Table 4 compares examples at their original evidence boundaries. It is deliberately not an accuracy leaderboard: some entries concern clock agreement, another a bounded visual check, and another an internal clock-translation statistic. Different hardware, reference instrumentation and observation durations preclude a pooled estimate.

Table 4. Representative results and their limits. Values are reported by the cited authors unless explicitly labeled calculated. SD denotes standard deviation. A signed mean close to zero does not bound individual errors.

Study and boundary	Reported evidence	Conditions and interpretation
RBS [20], receiver clock agreement	1.85 ± 1.28 microseconds (mean ± SD), using kernel timestamps	Commodity 802.11 clock-agreement experiment; not camera acquisition accuracy
Glossy [24], inferred common reference time	Average error about 0.4 microseconds through eight hops; SD reaches 4.8 microseconds at eight hops	Figure 13; 4000 measurements per setting; the small mean does not imply a sub-microsecond bound
TICSync [32], mapped clock time	Typically better than millisecond accuracy within seconds; probabilistic analysis accompanies the estimator	Abstract and evaluation; mapping accuracy depends on clock and delay assumptions
VersaVIS [27], camera acquisition check	Better than 0.5 ms in a 1 kHz LED-board test; 400 image pairs checked	Section 3.1; test resolution and observation count restrict the conclusion
VersaVIS [27], calibration and clock mapping	Camera-IMU offset consistency below 0.05 ms; host mapping reported as ±0.2 ms	Sections 3.2-3.3; calibration consistency and offset innovation are not independent absolute event-time truth
The Matter of Time [28], FPGA timestamping and LiDAR frames	Camera/IMU timestamping described as cycle-level; PTP LiDAR-frame variation within 100 microseconds	Section IV; separate sensor interfaces and observables; no common optical-event bound established
5G transparent clock [25], master-slave counter measurement	Test 1: mean -21.13 ns, SD 61.83 ns, minimum -179.89 ns, maximum 124.89 ns	20 minutes, 1 packet/s per listed message type, PTP-only traffic; calculated peak-to-peak span 304.78 ns
Cross-vendor gPTP [9], interoperability	Three classes of failure associated with delay calculation, clock control and traffic-class assignment	Engineering failure analysis; no single acquisition accuracy statistic is claimed

Three comparisons are especially informative. First, Glossy's eight-hop example shows why an average error must be interpreted together with its dispersion and does not establish a maximum-error bound. Second, VersaVIS reports several numbers for

different portions of one system. Its LED-based camera test, repeated camera-IMU calibration and host translation evaluation cannot be collapsed into one "synchronization accuracy". The host experiment's innovation statistic depends on its estimator and noise interpretation; an independent reference would be needed to establish external mapping accuracy. Third, the transparent-clock study measures a network-clock quantity with a precision counter, but does not measure sensor acquisition events. Its strongest result therefore applies to the timed clock endpoints and tested traffic condition.

The same reasoning limits software-versus-hardware comparisons. Hardware can improve observability at a specific boundary; software can estimate a clock relation or a physical offset using information unavailable to a network interface. An accurate software estimate under a favorable delay distribution does not remove the asymmetry ambiguity of Equation (2). Conversely, hardware timing cannot correct unknown exposure semantics solely by reducing packet-timestamp noise. Competitive learning for PTP delay distributions is a recent example of estimator research evaluated by simulation [39]. It motivates testing under uncertain delay models, while providing no basis for a universal deployed-robot accuracy claim.

7. Drift, Synchronization Intervals and Communication Cost

7.1. A conditional timing budget

After loss of a reference, a useful conservative model for a relative timing bound is

$$U(T) \leq U_0 + \rho T + \frac{1}{2}\kappa T^2. \quad (5)$$

U_0 is an initial bound in seconds, and ρ bounds residual relative frequency error at the last valid update. The parameter κ bounds its rate of change in inverse seconds. Equation (5) follows by integrating a bounded frequency error; it is an illustrative engineering model, not a guarantee for an uncharacterized oscillator. U_0 must include the non-growing contributions assigned to the same pairwise budget. If drift is approximated as constant, the longest interval consistent with target U_{req} is (U_{req} - U_0)/ρ, provided U_{req} exceeds U_0 and ρ is positive. Faster updates cannot resolve a budget already exhausted by fixed biases.

Consider a calculated example in which U_{req} is 100 microseconds and U_0 is 20 microseconds. Residual relative frequency error of 20 ppm permits an interval no longer than 4 s under the constant-drift approximation; 1 ppm permits 80 s. These are assumed residual errors after estimation, not specifications of a sensor. If the mapping or event-timing contribution alone exceeds the target, improving network synchronization cannot recover the missing budget. Figure 2 illustrates this sensitivity and its relation to protocol traffic.

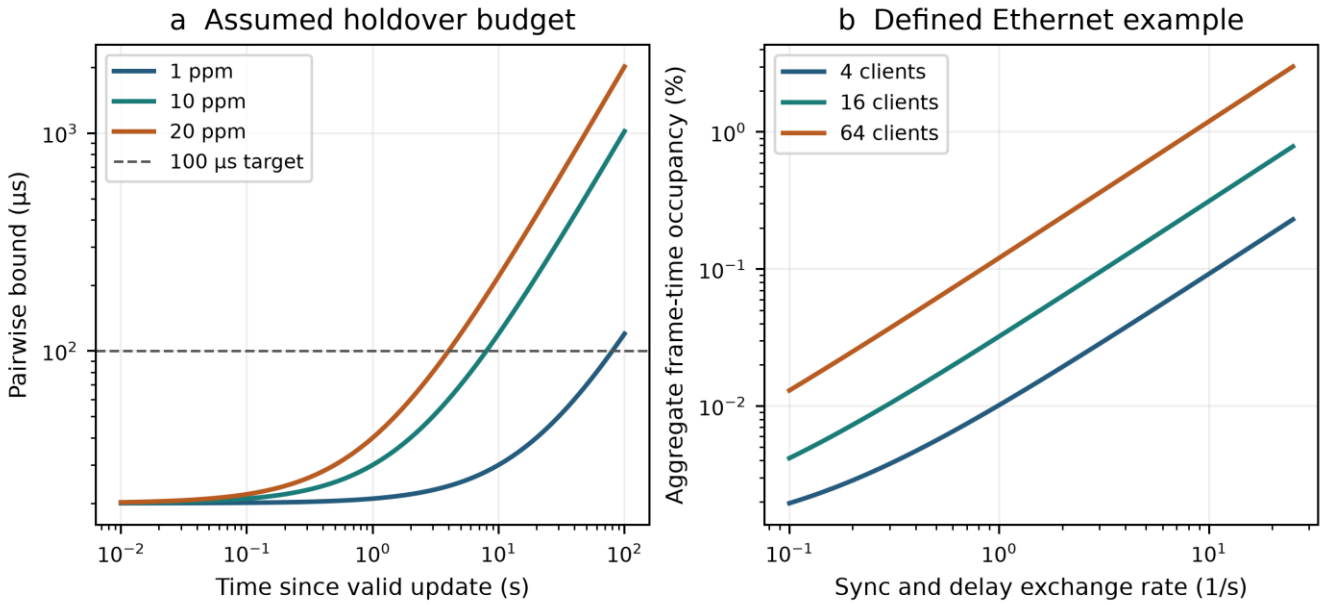


Figure 2. Illustrative calculations, not experimental results. Left: growth of a pairwise timing bound with U_0 = 20 microseconds, κ = 0 and assumed residual relative frequency errors of 1, 10 and 20 ppm. The horizontal line marks a 100-microsecond target. Right: aggregate Ethernet frame-time occupancy for the explicitly defined two-step end-to-end example with 4, 16 and 64 clients. Only the listed timing messages are counted. Curves do not estimate a particular protocol implementation's measured performance.

A timing target can also be related to application geometry. For small time error, a near-center lateral translation gives $|\delta_u| \approx f_{px}\ (|v_\perp|/Z)\ |\delta_t|$. A selected rotational component gives $|\delta_u| \approx f_{px}\ |\omega|\ |\delta_t|$ under the corresponding local approximation. These first-order approximations omit depth variation, rolling shutter, blur and estimator feedback. With assumed f_{px} = 500 pixels, Z = 2 m and lateral speed 1 m/s, a 0.5-pixel tolerance gives 2 ms. With the same focal length and angular speed 2 rad/s, it gives 0.5 ms. If both components act together, their combined image-motion Jacobian must be used; satisfying each isolated calculation is insufficient. Such calculations help allocate a requirement, but cannot establish a universal synchronization threshold for robots.

7.2. Accounting for messages and occupied time

Communication overhead depends on the transport, timing topology and accounting boundary. For an intentionally simplified two-step, multicast-Sync, end-to-end PTP example, let N clients share Sync, Follow_Up and Announce messages, while each client exchanges Delay_Req and Delay_Resp. If their frequencies are f_s, f_a and f_d, aggregate transmission count is

$$M_{E2E} = 2f_s + f_a + 2Nf_d. \quad (6)$$

Equation (6) counts transmissions in a single shared segment, excluding switch replication, management, signaling, retries and startup. It is not a mandatory message schedule. In a multi-link network, the relevant occupied time must be counted on each link. For two-step peer-delay exchanges, three messages per initiating endpoint per exchange give a term 3P f_p, where P is the number of endpoints initiating peer-delay measurement. If both ends of every bidirectional link initiate, P is twice the link count. This distinction avoids confusing link count with initiating-port count.

To illustrate byte accounting, assume untagged Ethernet, UDP/IPv4 and basic PTP messages without optional suffixes. The 34-byte PTP header gives 44-byte Sync, Follow_Up and Delay_Req messages, 54-byte Delay_Resp messages and 64-byte Announce messages [40]. With Ethernet header and frame check sequence, preamble and inter-frame

gap included as link-time occupancy, Sync, Follow_Up and Delay_Req occupy 110 byte-times each, Delay_Resp occupies 120, and Announce occupies 130. Thus,

$$B = 220f_s + 230Nf_d + 130f_a \text{byte-times/s.} \quad (7)$$

At N = 16, f_s = f_d = 8/s and f_a = 1/s, the calculated total is 31,330 byte-times/s, or 250,640 bit-times/s. Normalizing this aggregate to a nominal 100 Mbit/s rate gives 0.25064%. The normalization is a scalar cost comparison; a switched, full-duplex network requires separate accounting for each link and direction. It is not a measured utilization or a contention estimate. IPv6, VLAN tags, authentication, one-step operation, unicast scheduling or different intervals change the result. On Wi-Fi, a meaningful model must also include contention, acknowledgments, backoff, retries and PHY rate; Ethernet byte occupancy should not be reused as wireless airtime.

Table 5. Cost dimensions and their interpretation. Scaling statements refer to the explicitly stated topology and accounting model, rather than universal implementation complexity.

Mechanism	Cost that must be counted	Main qualification
Shared two-step end-to-end exchange	Shared Sync/Follow_Up plus per-client delay exchange	Equation (6); count replication per link
Peer-delay exchange	Exchanges per initiating port and adjacent link	Both endpoint directions may initiate
Receiver-reference broadcasts	Beacons, receiver timestamp sharing and listening	One beacon can serve several receivers; follow-on exchange still costs resources
Flooding	Forwarding, retransmission and constrained radio intervals	Reference age and topology affect useful accuracy
Trigger wiring	Timer channels, wiring, interface power and maintenance	Little timing-message traffic does not mean zero system cost
Temporal calibration	Compute, data windows and sufficient excitation	Can operate without dedicated timing packets; needs informative sensor data

Equations (5)-(7) expose a trade-off rather than an optimizer. Increasing update rate raises traffic while reducing the interval over which residual rate error accumulates, but it may also perturb queueing and estimator behavior. The 5G transparent-clock experiment found different behavior at different PTP rates in its specific setup [25]. A sensible design first determines whether error is dominated by update age, timestamp noise or a fixed bias, then changes the mechanism that addresses that component. A claim of energy efficiency additionally needs measurement or a justified power model; message counts alone are insufficient.

8. Selecting and Verifying an Architecture

8.1. Single-robot and distributed sensing

For a compact sensor head, a common timer with hardware event capture or triggering can make the local acquisition relation easier to observe. The connection from that timer to host and network time must still be established if other devices participate in fusion. For an Ethernet sensor network, hardware-timestamped PTP or gPTP can distribute a reference, while camera firmware schedules exposure against it or a gateway converts the reference into trigger signals. Basler documentation makes this separation explicit: PTP synchronizes clocks and Synchronous Free Run configures the acquisition schedule [41,42]. Feature support and configuration remain model dependent.

For a distributed team of robots, local acquisition timing and inter-robot clock transfer should be assessed separately. A tightly synchronized local sensor suite can continue to provide useful relative measurements during a network outage, while cross-robot fusion may require a growing uncertainty bound or temporary suspension. An application that merely orders access to a shared workspace has different requirements. Celik et al.'s 5G collaborative-robot study concerns distributed coordination and causal ordering [43]. Its access-time results should not be interpreted as errors in physical clock synchronization.

Figure 3 shows three representative arrangements. None is inherently optimal. The choice depends on event access, achievable calibration, network topology and the evidence needed by the consumer. A design using software translation can be appropriate when native sensor timestamps are reliable and timing requirements are moderate. A system with stringent cross-sensor phase requirements needs explicit event scheduling or capture, regardless of how well its network clocks agree.

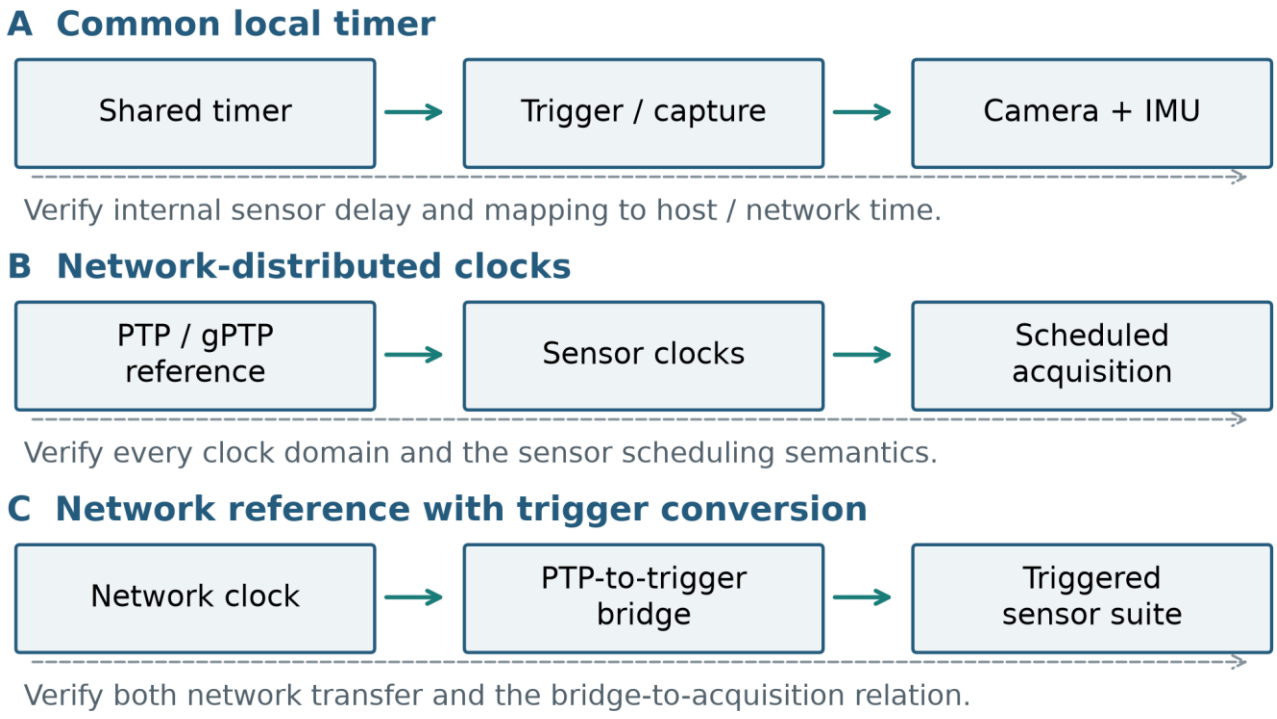


Figure 3. Three conceptual timing architectures and the boundaries that remain to be verified. A shared local timer, network-distributed clocks and a PTP-to-trigger bridge solve different integration problems. Dashed data paths emphasize that transport timing and acquisition timing need not coincide. The arrangements are schematic design examples, not measured implementations.

8.2. What public datasets contribute

Public robotic datasets are valuable because their documentation reveals real acquisition arrangements and provides sensor streams suitable for studying estimator sensitivity. EuRoC is an established visual-inertial benchmark [44]. TUM VI documents hardware synchronization of cameras and IMU [45]. The earlier Hilti platform integrates FPGA-managed sensor timing with PTP [46]. The 2023 challenge's robot instead uses PTP for LiDAR and a PTP-to-trigger board for cameras and IMU [47]. The latter documentation also states that stereo pairs independently control exposure [48]. That design makes exposure-start alignment and midpoint alignment a concrete distinction.

These resources do not automatically constitute synchronization benchmarks. Unless accompanied by network timestamps, oscillator-state logs, load traces and an independent reference, a sensor dataset cannot identify which network impairment caused a temporal inconsistency. An artificial timestamp shift added to a dataset tests an estimator's response to that perturbation. It does not reproduce physical clock drift, camera exposure changes or network congestion. This review therefore uses datasets as documented design examples and possible evaluation resources, without claiming a new benchmark experiment.

Table 6. Public resources and the claims they can support. Limitations refer to their use in this review; absence of a verified trace is not asserted as proof that no such trace exists elsewhere.

Resource	Useful evidence	Limit for network-timing evaluation
EuRoC [44]	Public visual-inertial benchmark and calibration context	No network load or holdover trace was verified here
TUM VI [45]	Hardware timing arrangement and calibrated sensor streams	Acquisition design does not validate another network deployment
Hilti 2022 [46]	Mixed sensor interfaces and documented synchronization checks	Platform-specific hardware and calibration
Hilti 2023 challenge [47,48]	PTP plus trigger conversion on a construction robot	Independent exposures and no verified concurrent network-load timing trace
LinuxPTP and kernel interfaces [11,12,13]	Access to timing configuration and clock observations	Operational logs need an independent reference for external accuracy

8.3. A minimum verification record

A convincing evaluation should state the application timing target, identify all clock domains, and record what physical event each timestamp denotes. Network-clock validation and sensor-event validation should use measurements appropriate to those boundaries. An external counter or oscilloscope can compare clock outputs or trigger lines, while a visual timing target or documented electrical event can test the relation between trigger and actual exposure. The instrumentation's resolution, uncertainty and observation duration need to be reported. Electrical edge alignment alone is insufficient evidence for optical midpoint alignment.

Verification should include steady state, startup, loss of reference and recovery, with background traffic and host load varied where relevant. Clock resets, missed triggers, duplicated frames and changing exposure or filter settings are particularly informative disturbances. The purpose is not to demand every stress test for every application, but to make the evaluated operating envelope explicit. Table 7 provides the minimum information needed to interpret a claim across implementations.

Table 7. Proposed reporting checklist for networked robotic sensor timing. This checklist is a synthesis for reproducible engineering reporting and has not been formally validated as a standard.

Item	Information to report	Why it matters
Requirement	Relative or external time target; acquisition or delivery deadline	Prevents comparison of different objectives
Timing chain	Reference, PHCs, MCU and host clocks; mapping direction	Reveals unverified clock boundaries
Timestamp semantics	Exposure start/midpoint, IMU sampling/output, LiDAR point/scan	Identifies the physical quantity being timed

Item	Information to report	Why it matters
Protocol configuration	Edition/profile, delay mechanism, intervals, topology	Defines communication and interoperability assumptions
Instrumentation	Independent reference, uncertainty, sample count and duration	Separates measured accuracy from servo residual
Error statistics	Signed bias, spread, percentiles or observed maxima with definition	Prevents a small mean from concealing large deviations
Operating conditions	Sensor modes, host/network load, temperature where relevant	Delimits transferability
Failure and cost	Reference loss/recovery, resets, valid-time flag, traffic and power basis	Exposes transient behavior and resource trade-offs

9. Discussion and Research Priorities

9.1. The main synthesis and its limits

The reviewed evidence supports a layered responsibility for time. Packet synchronization constrains clocks at measured network boundaries; local mapping extends that relation to other clocks; event scheduling or capture links time to sensing; calibration reconciles the remaining measurement-model offset. This synthesis explains why replacing software timestamps with hardware timestamps can improve one part of a system without establishing end-to-end acquisition accuracy. It also explains why temporal calibration can improve fusion while leaving the network's absolute time uncertainty unresolved.

The strength of that conclusion is architectural rather than statistical. The examples are heterogeneous and were selected for explanatory value. Some sources were available only through abstracts or indexed primary text; implementation documentation is subject to change. Older radio experiments demonstrate mechanisms but do not characterize current camera networks. Access to public standards summaries does not replace examination of all normative clauses. No pooled performance effect, coverage-completeness claim or experimentally validated decision rule follows from this corpus. These limits are reflected in the evidence ledger and in the conditional wording of the design guidance.

9.2. Timing validity during degraded operation

The next useful advance is to expose timing validity in the sensor data path. A consumer needs to know when the last valid mapping was established, which reference and timescale were used, and whether a reset or discontinuity invalidated the relation. An uncertainty estimate or conservative interval should age during holdover according to an identified model. A binary “locked” state is useful operationally but cannot by itself convey exposure semantics or the remaining pairwise error budget. The proposal is a design recommendation derived from the timing chain, not an experimentally verified metadata standard.

Cross-vendor gPTP failures show why recovery behavior belongs in that interface [9]. A clock jump may affect neighboring estimators and applications differently. Recording clock identity, mapping version and validity interval would make it easier to detect data spanning a discontinuity. Such metadata could also help distributed robot learning pipelines preserve the relationship between observations and actions. The evidence reviewed here does not quantify a downstream embodied-AI benefit; that would require a separate study with controlled timing perturbations and task outcomes.

9.3. Research that resolves the remaining evidence gaps

Three research directions follow directly from the unresolved boundaries. First, acquisition-aware benchmarks should combine packet traces, clock-state logs and

independently observed sensor events. A shared multimodal target observed by heterogeneous sensors, together with instrumented timing paths, could separate network error from acquisition and interpretation error. The benchmark must document its own timing uncertainty so that the reference does not become the dominant unknown.

Second, adaptive synchronization should optimize a declared acquisition budget and communication cost, rather than minimize an internal residual alone. Equation (5) indicates what needs to be identified: residual relative rate, its variation and the fixed part of the budget. An adaptive policy should be compared at matched timing violation rates and matched traffic or energy accounting boundaries. Learned delay models are promising, but require checks against changing topology, sensor modes and unseen delay distributions [39]. Network congestion information may improve filtering while leaving constant asymmetry and sensor delays to other mechanisms [26].

Third, timing integrity should include security. RFC 7384 describes threats to time protocols, including manipulation of message delay [49]. RFC 8915 specifies Network Time Security for NTP [50]. Authentication helps establish message origin and integrity, but does not prove an unknown one-way propagation delay. A robot timing design should therefore separate trust in the reference from confidence in transfer delay and event mapping. Redundant observations and explicit degradation policies are research opportunities; their value must be demonstrated for a stated failure model rather than asserted from redundancy alone.

10. Conclusions

Clock synchronization for networked robotic sensing is best assessed by tracing time from a reference to a physical acquisition event. The central distinction is between clock agreement, event alignment and the interpretation of the delivered timestamp. Hardware timestamping, packet-delay compensation, shared triggers, clock translation and temporal calibration address different parts of that chain and can be combined when their assumptions are explicit.

Published performance values support conditional comparisons when measurement boundaries, statistics and operating conditions are retained. They do not support a universal protocol accuracy ranking. The proposed error accounting and illustrative drift/traffic calculations provide a way to connect application requirements to synchronization design, while the reporting checklist identifies the observations needed to test that design. For heterogeneous robot sensor networks, the most consequential next step is reproducible validation across network, local-clock and acquisition boundaries, including the periods when synchronization is lost and recovered.

Supplementary Materials: Supplementary File S1 contains the search and evidence methodology, extended claim audit and calculation definitions. Machine-readable evidence tables, verified bibliographic records and the code for the illustrative calculations and figures accompany the submission package. Third-party papers remain at their original source URLs and are not redistributed as supplementary content.

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Abbreviations

Abbreviation	Definition
gPTP	generalized Precision Time Protocol
FPGA	field-programmable gate array
IMU	inertial measurement unit
LiDAR	light detection and ranging
MCU	microcontroller unit
NIC	network interface controller
NTP	Network Time Protocol
PHC	PTP hardware clock
PTP	Precision Time Protocol
RBS	Reference-Broadcast Synchronization
SD	standard deviation
SLAM	simultaneous localization and mapping
TSN	Time-Sensitive Networking
WSN	wireless sensor network

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